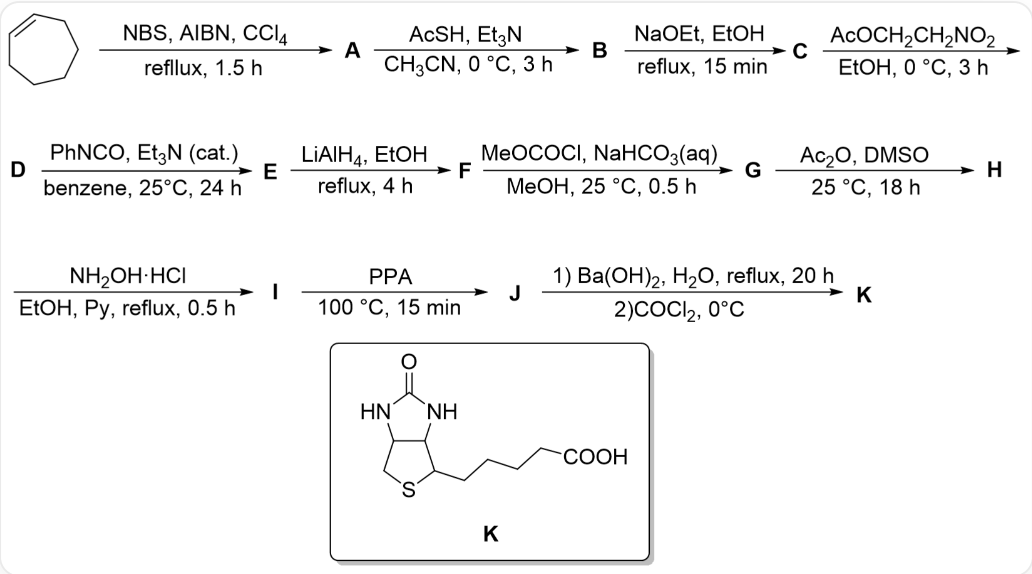
Question

The following is a total synthesis route:



Compound $\mathbf{K}$ (SMILES O=C(N1)NC2C1CSC2CCCCC(O)=O) was synthesized through a multi-step reaction: C1=CCCCCC1>>[A], [A]>>[B], [B] >>[C], [C]>>[D], [D]>>[E], [E]>>[F], [F]>>[G], [G]>>[H], [H]>>[I], [I]>>[J], [J]>>[K]. The reaction conditions are: (1) $\mathrm{NBS, AIBN, CCl_4}$, reflux for 3 hours (2) $\mathrm{AcSH, Et_3N, CH_3CN,}$, react at 0 degrees Celsius for 4 hours (3) $\mathrm{EtONa, EtOH}$, reflux for 15 minutes (4) $\mathrm{AcOCH_2CH_2NO_2, EtOH}$, react at 0 degrees Celsius for 3 hours (5) $\mathrm{PhNCO, Et_3N}$(catalytic amount), PhH, react at 25 degrees Celsius for 24 hours (6) $\mathrm{LiAlH_4, EtOH}$, reflux for 4 hours (7) $\mathrm{MeOCOCl, NaHCO_3(aq), MeOH}$, react at 25 degrees Celsius for half an hour (8) $\mathrm{Ac_2O, DMSO}$, react at 25 degrees Celsius for 18 hours (9) $\mathrm{NH_2OH \cdot HCl, EtOH, Py}$, reflux for half an hour (10) PPA, react at 100 degrees Celsius for 15 minutes (11) Divided into two steps, first $\mathrm{Ba(OH)_2, H_2O}$, reflux for 20 hours, and then add $\mathrm{COCl_2}$ and react at 0 degrees Celsius

Given that the molecular formula of $\mathbf{D}$ is $\mathrm{C_9H_{15}NO_2S}$, and $\mathbf{C}$ is a sodium salt. Which of the following statements is correct:

1. The conversion of $\mathbf{D}$ to $\mathbf{E}$ involves an intermediate containing a carbon-nitrogen triple bond.

2. $\mathbf{E}$ contains 2 rings.

3. The molecular formula of $\mathbf{H}$ is $\mathrm{C_{11}H_{17}NO_3S}$.

4. **J** contains a seven-membered ring.

5. The transformation from **I** to **J** may produce a compound with high ring strain, with a molecular formula of $C_{11}H_{16}N_2O_2S$.

A. All other options are incorrect

B. 2, 4

C. 1, 3, 5

D. 1, 2

E. 2, 4, 5

F. 1, 4, 5

G. 1, 2, 4, 5

H. 1, 2, 3, 4

I. 2, 3, 4

J. 3, 4, 5

K. 4

L. 3, 5

M. 1, 2, 3, 4, 5

N. 2, 5

Answer

Correct Answer: C

Detailed Explanation

(1) The reaction is allylic radical halogenation, $\mathbf{A}$ is BrC1C=CCCCC1

CHECKPOINT 1 PTS

$\mathbf{A}$ is BrC1C=CCCCC1

(2) The reaction is the substitution of $\mathrm{Br^-}$ by $\mathrm{AcS^-}$, $\mathbf{B}$ is CC(SC1C=CCCCC1)=O

CHECKPOINT 1 PTS

$\mathbf{B}$ is CC(SC1C=CCCCC1)=O

(3) The reaction is the removal of the acetyl group by $\mathrm{EtO^-}$, $\mathbf{C}$ is [Na]SC1C=CCCCC1

CHECKPOINT 1 PTS

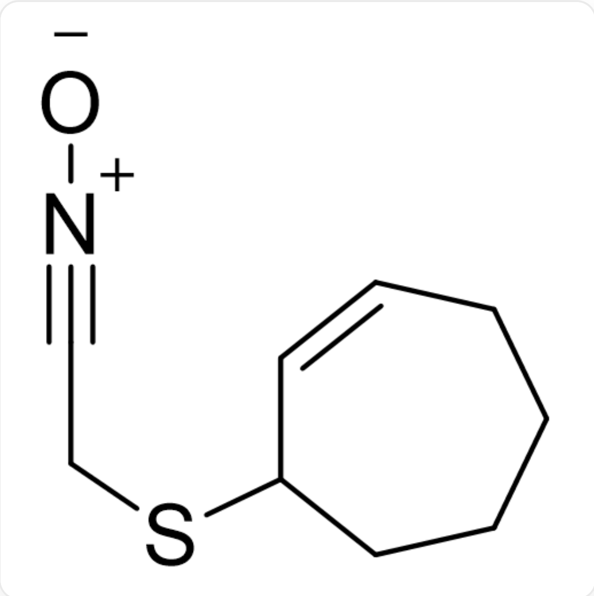
$\mathbf{C}$ is [Na]SC1C=CCCCC1

(4) Acetylation can occur here, but it is meaningless since the acetyl group was just removed in the previous step, so it can only be the substitution of AcO^{-}, $\mathbf{D}$ is O=[N+](CCSC1C=CCCCC1)[O-]

CHECKPOINT 1 PTS

$\mathbf{D}$ is O=[N+](CCSC1C=CCCCC1)[O-]

(5) This condition is very classic, it can convert nitro compounds to nitrile oxide [O-][N+]#CCSC1C=CCCCC1, which is relatively reactive, and subsequent intramolecular [3+2] reaction easily occurs to give $\mathbf{E}$: C12C3CCCCC1ON=C2CS3



SMILES is [O-][N+]#CCSC1C=CCCCC1

CHECKPOINT 1 PTS

$\mathbf{E}$ is C12C3CCCCC1ON=C2CS3

Therefore, the intermediate contains a carbon-nitrogen triple bond, and $\mathbf{E}$ contains three rings, statement 1 is correct, statement 2 is wrong

(6) $LiAlH_4$ reductively cleaves the N-O bond, and then reduces the carbon-nitrogen double bond to give $\mathbf{F}$: NC(CS1)C2C1CCCCC2O

CHECKPOINT 1 PTS

$\mathbf{F}$ is NC(CS1)C2C1CCCCC2O

(7) MeOCOCl reacts with the most nucleophilic amino group in $\mathbf{F}$ to give $\mathbf{G}$: OC1CCCCC2C1C(NC(OC)=O)CS2

CHECKPOINT 1 PTS

$\mathbf{G}$ is OC1CCCCC2C1C(NC(OC)=O)CS2

(8) If the hydroxyl group in $\mathbf{F}$ is acetylated, it is meaningless to add hydroxylamine later; therefore, DMSO has oxidizing properties, here it is similar to Swern oxidation, but the dehydrating agent is replaced by Ac_2O. $\mathbf{H}$: O=C1CCCCC2C1C(NC(OC)=O)CS2

CHECKPOINT 1 PTS

$\mathbf{H}$ is O=C1CCCCC2C1C(NC(OC)=O)CS2

The molecular formula of $\mathbf{H}$ is $C_{11}H_{17}NO_3S$, statement 3 is correct

(9) Hydroxylamine condenses with the carbonyl group to give $\mathbf{I}$: O/N=C1CCCCC2C\1C(NC(OC)=O)CS2

CHECKPOINT 1 PTS

$\mathbf{I}$ is O/N=C1CCCCC2C\1C(NC(OC)=O)CS2

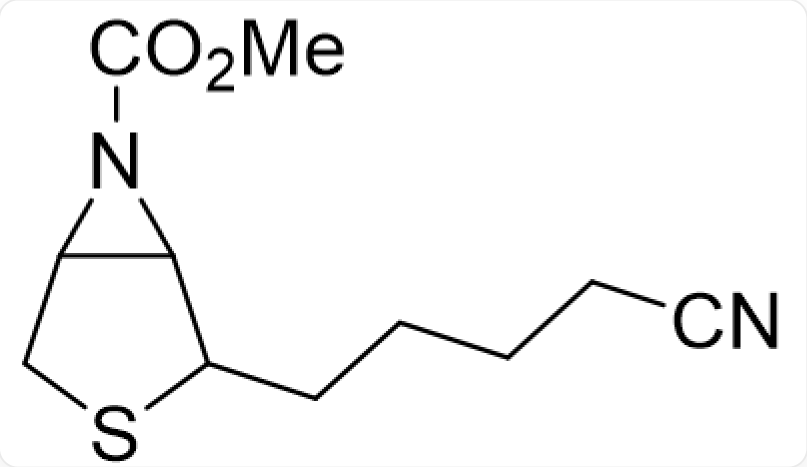
(10) Beckmann rearrangement occurs to give $\mathbf{J}$: O=C(N1)CCCCC2C1C(NC(OC)=O)CS2

CHECKPOINT 1 PTS

$\mathbf{J}$ is O=C(N1)CCCCC2C1C(NC(OC)=O)CS2

$\mathbf{J}$ contains a five-membered ring and an eight-membered ring, statement 4 is wrong

Beckmann rearrangement may also not involve alkyl migration, but rather cleavage of the C-C bond to form a carbocation (which can be stabilized by the neighboring sulfur atom), followed by closing the three-membered ring and dehydration to form O=C(N1C2C1CSC2CCCCC#N)OC, the molecular formula is $C_{11}H_{16}N_2O_2S$, statement 5 is correct

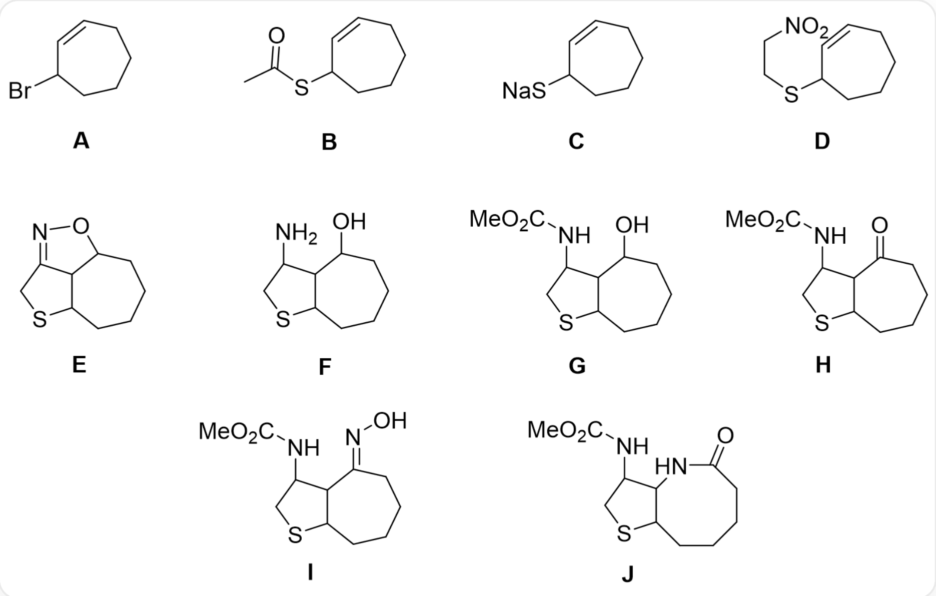


SMILES is O=C(N1C2C1CSC2CCCCC#N)OC

CHECKPOINT 1 PTS

Beckmann rearrangement can form O=C(N1C2C1CSC2CCCCC#N)OC byproduct

(11) Hydrolyze the amide, then react the two amino groups with $COCl_2$ to close the five-membered ring



The figures are $\mathbf{A}$~$\mathbf{J}$ respectively, see the above process for SMILES